INCH-POUND

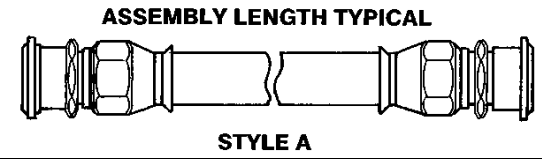
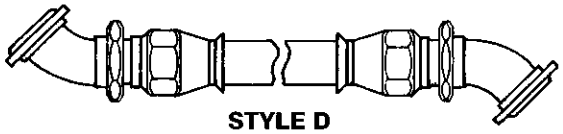
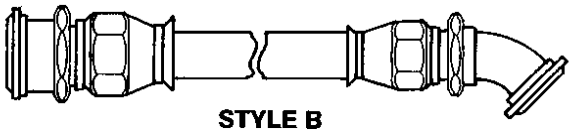
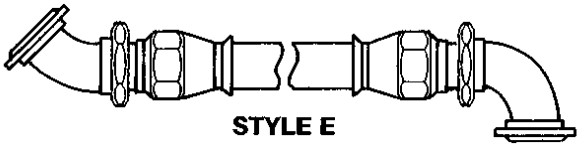
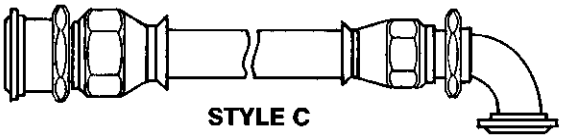
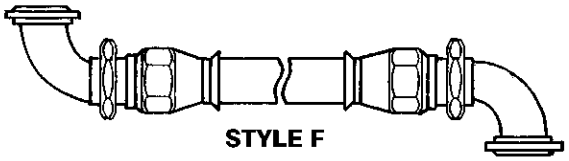
MIL-DTL-32434/5
23 January 2013

DETAIL SPECIFICATION SHEET

# HOSE ASSEMBLY, POLYTETRAFLUOROETHYLENE, CONVOLUTED, REUSABLE FITTINGS, HIGH TEMPERATURE, MEDIUM PRESSURE, FLANGE-TO-FLANGE

This specification is approved for use by all
Departments and Agencies of the Department of Defense.

The requirements for acquiring the product described herein shall consist of this specification sheet and MIL-DTL-32434.

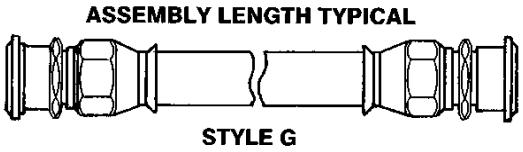
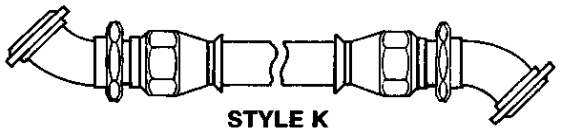
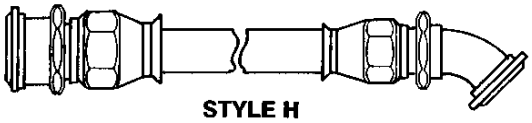
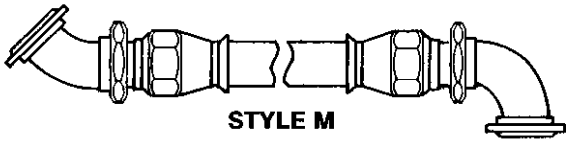
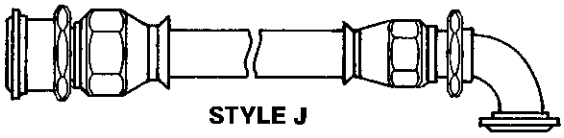
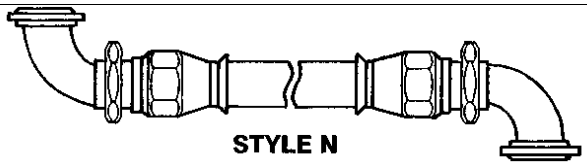
| Hose assembly with class 1 fittings (450°F) corrosion resistant steel | Hose assembly with class 1 fittings (450°F) corrosion resistant steel |
|---|---|
| End 1 End 2 | End 1 End 2 |
|   STYLE A |  STYLE D |
|  STYLE B |  STYLE E |
|  STYLE C |  STYLE F |

NOTES:
1. See table I for fitting ends and HCOF.
2. Fittings shall mate with the mounting pad as shown in MS33786.

FIGURE 1. Class 1 hose assembly styles with flange-to-flange reusable fittings.



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| Hose assembly styles with class 2 fittings (275°F) combination of aluminum and corrosion resistant steel | | Hose assembly styles with class 2 fittings (275°F) combination of aluminum and corrosion resistant steel | |
|---|---|---|---|
| End 1 | End 2 | End 1 | End 2 |
|  STYLE G | |  STYLE K | |
|  STYLE H | |  STYLE M | |
|  STYLE J | |  STYLE N | |

NOTES:
1. See table II for fitting ends and HCOF.
2. Fittings shall mate with the mounting pad as shown in MS33786.

FIGURE 2. Class 2 hose assembly styles with flange-to-flange reusable fittings.

Hose cut-off factor (HCOF) for assembly with class 1 fittings: See table I.



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TABLE I. Hose cut-off factor (HCOF) for assembly with class 1 fittings. 1/

| Hose assembly styles with class 1 corrosion resistant steel fittings (CRES) 2/ | Fitting ends | | HCOF (size vs length) 3/ | | | | | |
|---|---|---|---|---|---|---|---|---|
| | 1 | 2 | 8 | 10 | 12 | 16 | 20 | 24 |
| A | MS27054 | MS27054 | 2.46 | 2.84 | 2.76 | 3.22 | 3.38 | 3.62 |
| B | MS27054 | MS27056 | 2.48 | 2.84 | 3.28 | 3.59 | 3.91 | 4.31 |
| C | MS27054 | MS27058 | 2.44 | 2.83 | 3.30 | 3.66 | 4.03 | 4.49 |
| D | MS27056 | MS27056 | 2.50 | 2.84 | 3.80 | 3.96 | 4.44 | 5.00 |
| E | MS27056 | MS27058 | 2.46 | 2.83 | 3.82 | 4.03 | 4.56 | 5.18 |
| F | MS27058 | MS27058 | 2.42 | 2.82 | 3.84 | 4.10 | 4.58 | 5.36 |

1/ Class 1 fittings, styles A through F, are made from CRES.
2/ For depiction of styles see figure 1.
3/ The HCOF is used in the following calculation to determine the hose length required to produce an assembly of a specific size, style and length: Assembly length - HCOF = Hose length. For example, the hose length required to produce a 23.50 inch length assembly of size 10, style F is calculated as follows: 23.50 - 2.82 = 20.68.

HCOF for assembly with class 2 fittings: See table II.

TABLE II. HCOF for assembly with class 2 fittings. 1/

| Hose assembly styles with class 2 aluminum and CRES fittings 2/ | Fitting ends | | HCOF (size vs length) 3/ | | | | | |
|---|---|---|---|---|---|---|---|---|
| | 1 | 2 | 8 | 10 | 12 | 16 | 20 | 24 |
| G | MS27054 | MS27054 | 2.46 | 2.84 | 2.76 | 3.22 | 3.38 | 3.62 |
| H | MS27054 | MS27056 | 2.48 | 2.84 | 3.28 | 3.59 | 3.91 | 4.31 |
| J | MS27054 | MS27058 | 2.44 | 2.83 | 3.30 | 3.66 | 4.03 | 4.49 |
| K | MS27056 | MS27056 | 2.50 | 2.84 | 3.80 | 3.95 | 4.44 | 5.00 |
| M | MS27056 | MS27058 | 2.46 | 2.83 | 3.82 | 4.03 | 4.56 | 5.18 |
| N | MS27058 | MS27058 | 2.42 | 2.82 | 3.84 | 4.10 | 4.68 | 5.36 |

1/ Class 2 fittings, styles G through N, are made from combination of aluminum and CRES.
2/ See figure 1 for depiction of styles.
3/ The HCOF is used in the following calculation to determine the hose length required to produce an assembly of a specific size, style and length: Assembly length - HCOF = Hose length. For example, the hose length required to produce a 23.50 inch length assembly of size 10, style F is calculated as follows: 23.50 - 2.82 = 20.68.



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REQUIREMENTS

Assembly classification: Class 1 and class 2 hose assemblies, as specified in MIL-DTL-32434, have been incorporated into the Part or Identifying Number (PIN) as a part of styles (see tables I and II and figures 1 and 2).

Dimensions: Unless otherwise specified, all dimensions are in inches.

Fittings: All fittings shall be qualified in accordance with MIL-DTL-27272.

Hose: The hose shall be in accordance with MIL-DTL-32435.

Hose size and elbow drop height: Hose size code and elbow drop height see table III.

TABLE III. Size code and elbow fitting drop height.

| Size | Reference tube OD | Size code | Maximum drop height of elbow fitting | |
|---|---|---|---|---|
| | | | (a) 1/ | (b) 1/ |
| 8 | .500 | H | .458 | .927 |
| 10 | .625 | J | .410 | .931 |
| 12 | .750 | K | .503 | 1.191 |
| 16Z | 1.000 | M | .540 | 1.317 |
| 20Z | 1.250 | N | .605 | 1.535 |
| 24Z | 1.500 | P | .659 | 1.723 |

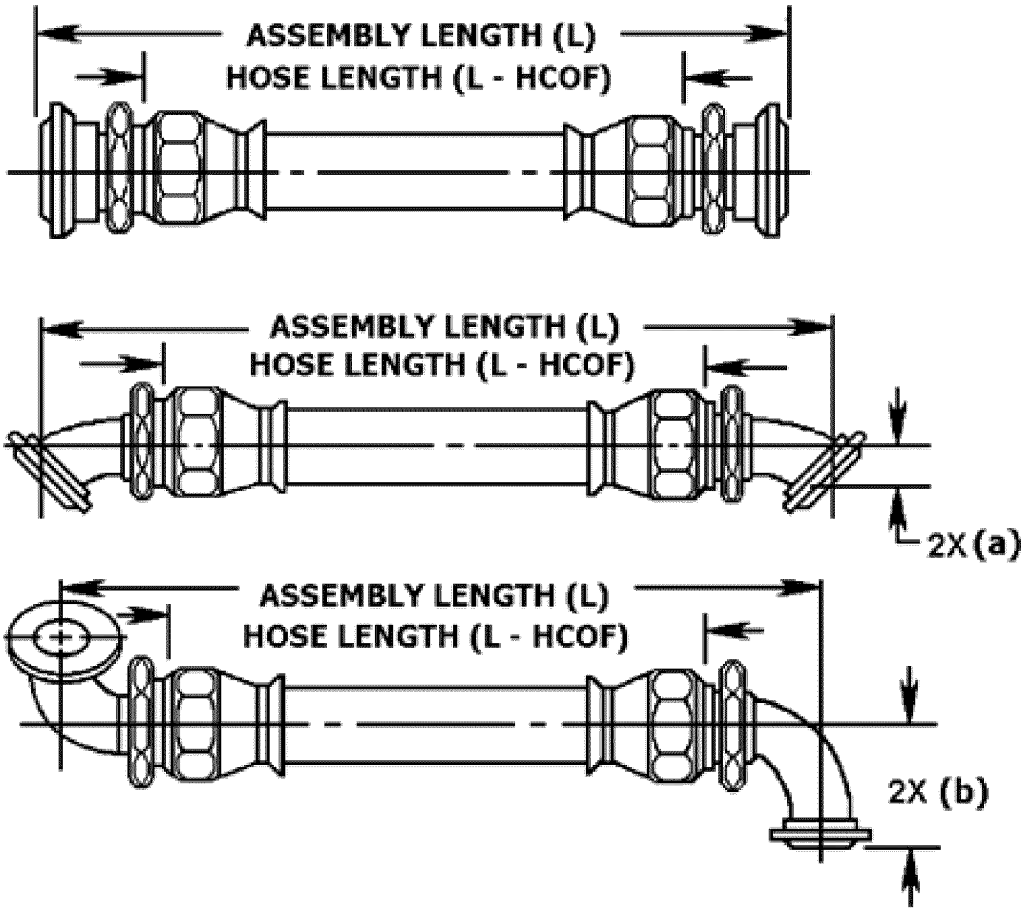
1/ Dimensions (a) and (b) are shown on figure 3.



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Assembly length and tolerance: Hose assembly shall be furnished in lengths as specified in the contract or purchase order (see figure 3); however, tolerances on the length of each hose assembly shall be as follows:

a. ± 1/8 inch for lengths under 18 inches.
b. ± 1/4 inch for lengths from 18 inches to 36 inches.
c. ± 1/2 inch for lengths from 36 inches to 50 inches.
d. ± 1% for lengths over 50 inches.



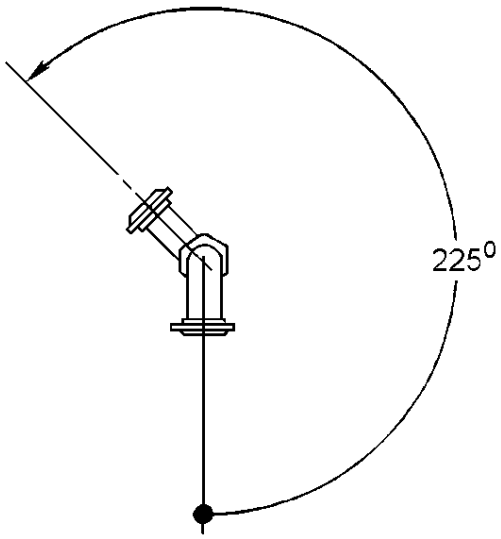

NOTES:
1. Hose assembly length “L” shall be measured, with the hose laid out horizontally and straight between the centers of the flange sealing surfaces, along a straight line parallel to the hose length.
2. See table III for dimensions of (a) and (b).

FIGURE 3. Elbow fitting drop height, dimensions (a) and (b)



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Angular alignment: Hose assemblies with elbow fittings on each end shall have the angular orientation between the elbows measured counter-clockwise from the centerline of the nearest fitting, positioned at six-o'clock, to the centerline of the other fitting (see figure 3). The elbow fitting drop height shall be as shown on figure 3 and table III. When applicable, the angular alignment shall be expressed in three digits and specified in the PIN.




NOTE: Angular alignment shall be measured in degrees with a tolerance of ±2°.

FIGURE 3. Measurement of angular alignment between elbow fittings.

Protective sleeve: If required, the hose assembly shall include a protective sleeve (see table IV) and its code shall be included in the PIN. Fire protective sleeve shall be subjected to testing in accordance with MIL-DTL-32434.



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TABLE IV. Protective sleeve code.

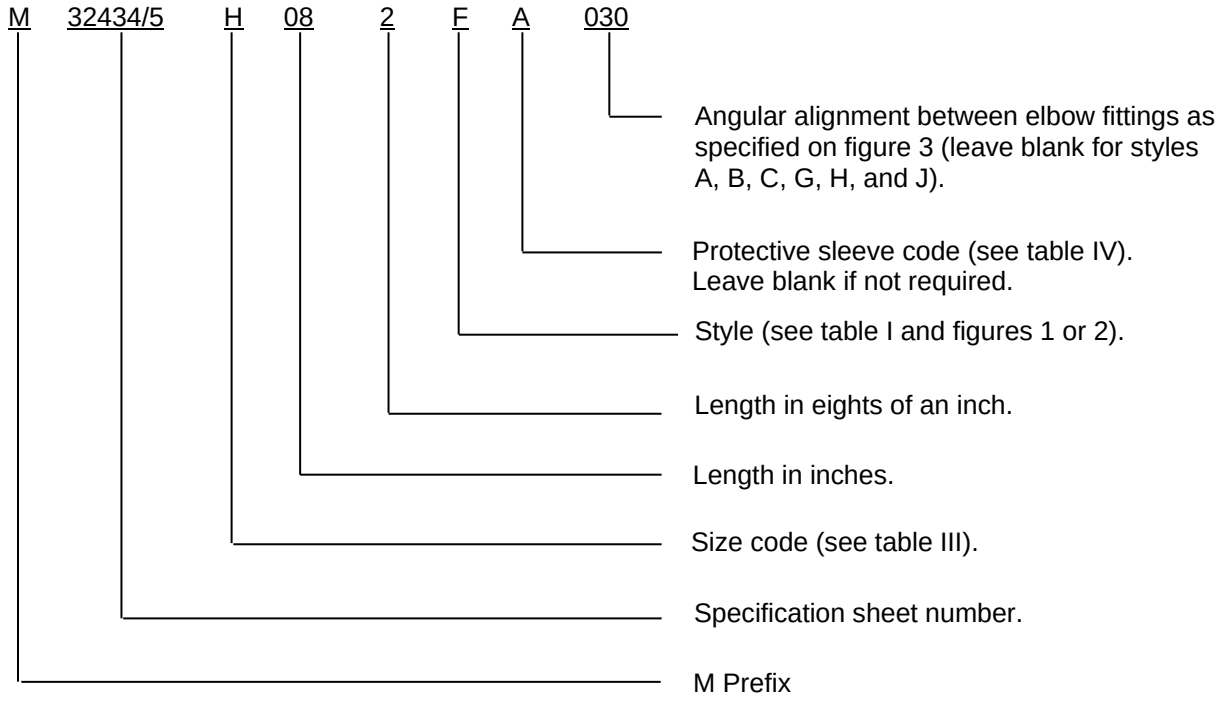
| Code | Type |
|---|---|
| Blank | None |
| A | SAE-AS1072 sleeve, fire protection, silicone covered, temperature ranging from -65°F to 450°F and intermittently to 500°F, secured with CRES bands as required. 1/ |
| B | SAE-AS1073 - code B sleeve, abrasion protection, heat shrinkable, black polyolefin, temperature ranging from -65°F to 250°F. |
| C | SAE-AS1291 - code A sleeve, chafe guard, extruded seamless white PTFE, temperature ranging from -65°F to 450°F, secured with CRES bands as required. |
| D | SAE-AS1291 - code C sleeve, chafe guard, extruded seamless transparent FEP, temperature ranging from -65°F to 350°F, secured with CRES bands as required. |
| E | SAE-AS1298 sleeve, heavy wall chafe guard, extruded seamless black PTFE, temperature ranging from -65°F to 450°F, secured with CRES bands as required. |
| L | Lock-wire hole |
| F | Code A + L |
| G | Code B + L |
| H | Code C + L |
| J | Code D + L |
| K | Code E + L |

1/ To prevent wicking of fluids, the cut end of the fire protective sleeve (code A) shall be coated with room temperature vulcanized (RTV) silicone rubber prior to installation. After installation, cracks and voids in the fire protective sleeve shall be coated with RTV rubber to prevent exposure of fiberglass.



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PIN: The PIN consists of letter M, specification sheet number, hose size code, hose length in inches and an eighths of an inch, hose assembly style, sleeve code, and angular alignment in degree between elbow fittings.




PIN example: M32434/5H082FA030 represents ½ inch OD PTFE convoluted hose assembly, 8.25 inches long, flange-to-flange reusable fitting with fire protective sleeve and 30O angular alignment between elbows.

Referenced documents. In addition to MIL-DTL-32434, this document references the following:

| | |
|---|---|
| MIL-DTL-27272 | MS33786 |
| MIL-DTL-32435 | SAE-AS1072 |
| MS27054 | SAE-AS1073 |
| MS27056 | SAE-AS1291 |
| MS27058 | SAE-AS1298 |



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CONCLUDING MATERIAL

Custodians:
Army - AV
Navy - AS
Air Force - 99
DLA - CC

Preparing activity:
DLA - CC

(Project 4720-2012-023)

Review activities:
Army - AR, AT, EA, MI
Navy - MC, SA, SH
Air Force - 70

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